Renewal Bandits

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Non-monotone $p_N^\dagger(t)$ (example)

We provide a numerical example in which the upper threshold $p_N^\dagger(t)$ is not strictly increasing.

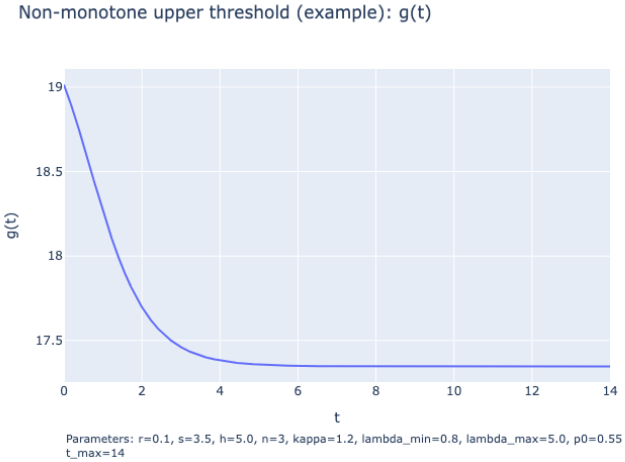
Hazard rate:

$$\lambda(t) = (1 - e^{-\kappa t})\lambda_{\min} + e^{-\kappa t}\lambda_{\max}.$$

Parameter values

- $r = 0.1$
- $s = 3.5$
- $h = 5.0$
- $N = 3$
- $\kappa = 1.2$
- $\lambda_{\text{min}} = 0.8$
- $\lambda_{\text{max}} = 5.0$
- $p_0 = 0.55$

$g(t)$

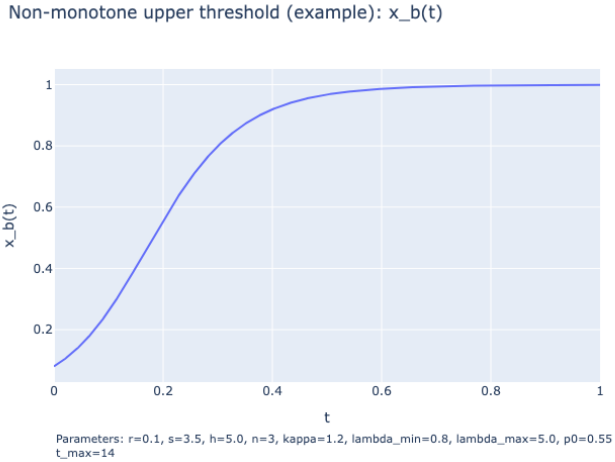

Non-monotone upper threshold (example): g(t)
19
18.5
18
17.5
g(t)
0
2
4
6
8
10
12
14
t
Parameters: r=0.1, s=3.5, h=5.0, n=3, kappa=1.2, lambda_min=0.8, lambda_max=5.0, p0=0.55
t_max=14

Single-crossing condition

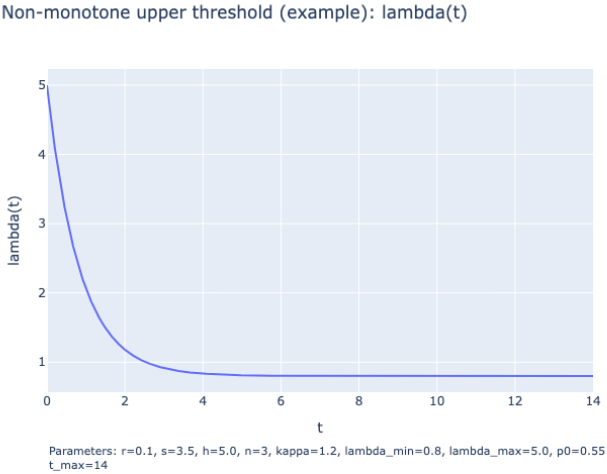
- The following ratio is strictly increasing, hence we do not have a violation of the single-crossing condition:

$$x_b := \frac{p_N^\dagger(t)}{p_N^\dagger(t) + (1 - p_N^\dagger(t))e^{-N\Lambda(t)}}$$

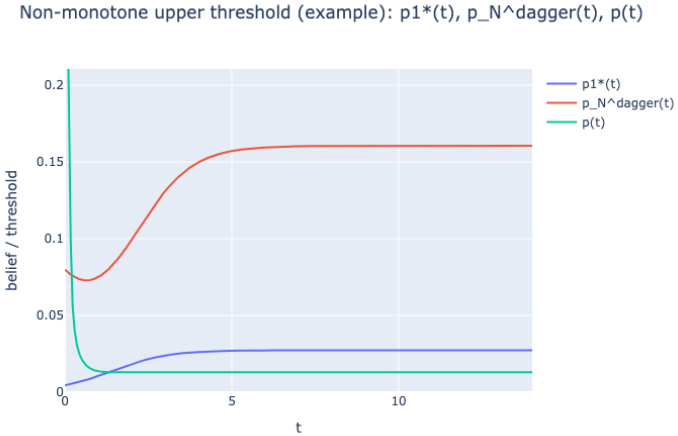
Single-crossing condition



Single-crossing condition

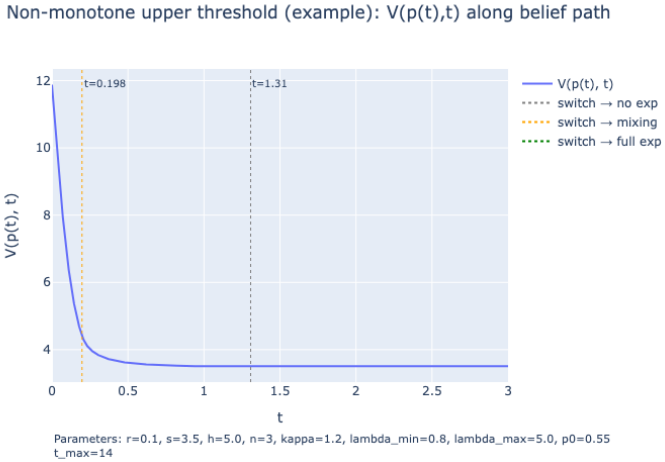


Thresholds and beliefs



Parameters: r=0.1, s=3.5, h=5.0, n=3, kappa=1.2, lambda_min=0.8, lambda_max=5.0, p0=0.55
t_max=14

Value along the belief path



Thank you.